

Multiple Choice (10 marks)

1. How is iron transported in the circulation from the intestine to the sites of metabolism in the body?
 - (a) As simple Fe^{2+} in the serum
 - (b) Bound to albumin
 - (c) Bound to ferritin
 - (d) Bound to transferrin
2. How much dietary protein should athletes consume daily?
 - (a) <0.8 g protein per kg bodyweight per day
 - (b) $0.8\text{-}1.2$ g protein per kg bodyweight per day
 - (c) $1.2\text{-}1.7$ g protein per kg bodyweight per day
 - (d) >2.0 g protein per kg bodyweight per day
3. Which of the following statements is not true?
 - (a) Docosahexaenoic acid can be synthesised from linolenic acid in vegans
 - (b) Docosahexaenoic acid is absent from vegan diets
 - (c) Lack of docosahexaenoic acid in vegans causes visual and cognitive impairment
 - (d) Microalgae can be used to synthesise docosahexaenoic acid
4. Deficiency of which of the following vitamins results in megaloblastic anaemia?
 - (a) Folate
 - (b) Niacin
 - (c) Riboflavin
 - (d) Vitamin B_6
5. Ulcerative Colitis
 - (a) Carries a risk of colonic cancer
 - (b) Can be treated with a LOFFLEX diet
 - (c) Can be caused by small intestinal bacterial overgrowth
 - (d) Can be caused by milk intolerance

True / False (10 marks)

Answer True or False

6. True or False: The prevalence of alcoholic myopathy in alcoholics is estimated to be between 40 to 60%.
7. True or False: Intestinal failure may benefit from sodium supplementation of 90 mmol/l or more.
8. True or False: Muscle glycogen is the primary fuel source for muscle contraction during short-term intense activities like sprinting.
9. True or False: Only curcumin has epigenetic effects among diet-derived compounds.
10. True or False: The worldwide prevalence of obesity is around 39%, indicating a severe global health crisis.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how malnutrition in older adults impacts their health, particularly in relation to mortality and morbidity. Provide examples to support your analysis.
12. Discuss the mechanisms by which non-esterified fatty acids are taken up by cells, including the roles of diffusion and carrier-mediated transport.

End of Paper